

Murali Sai Buddakkagari Venkata

AI Engineer

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EXPERIENCE

Dreamline AI – AI Engineer

Sep 2026 – Present

Community Dreams Foundation · Remote

- Designed a Supabase/PostgreSQL schema extension that lets a 141-program incentive catalog pre-fill homeowner applications, with a 21-key requirement registry, a three-valued JSON condition engine, and a readiness API.
- Secured it with Row Level Security, integrity constraints and audit triggers, passing 7/7 acceptance criteria locally with 0 of 141 records altered in an idempotent, rollback-ready release.
- Adapted the migration to the live schema in a Lovable planning pass (reused existing columns, separated eligibility from requirements, fixed an audit-trigger blocker) after baselining production data; SQL submitted for review.

Querkey Inc – Data Science Intern

Jun 2025 – May 2026

CA, US

- Built and deployed a proof-of-concept real-time Patient Flow Tracking System using multimodal AI (Gemini 2.5 Flash), chain-of-thought and few-shot prompting, and RAG to structure patient notes for clinical insight retrieval.
- Cut production ML workflow failures 60% across 100K+ records with a Dockerized observability stack (MLflow, FastAPI, ChromaDB, HuggingFace monitoring) enabling real-time failure detection.
- Authored a 30-query SQL bank to ground an AI chatbot's RAG answers in hospital data, and built a Superset REST API integration that renders dashboard charts on demand.

View Group India – AI and Automation Intern

Jan 2024 – Aug 2024

Formerly View Synergy · KA, IND

- On a cross-functional team modernizing a live platform, reviewed React, .NET Azure Functions, and MySQL code, root-causing 2 recurring exceptions and surfacing SQL-injection vulnerabilities in a 94-issue security assessment.
- Built and tested a new Roles/Designations CRUD feature end to end (MySQL stored procedures, .NET, React).
- Audited Redis usage across 6 backend microservices and documented Microsoft Entra ID app registration and role setup, supporting a proposal to replace a custom JWT/BCrypt authentication system.

PROJECTS

SEC EDGAR RAG Filing Analyzer

- Architected and deployed a production RAG system over 8,232 SEC 10-K chunks across 5 companies using FastAPI, ChromaDB, cross-encoder reranking and RBAC information barriers (FINRA 2241) on Cloud Run, verified by a 504-test CI suite and a 54-question eval (0.853 faithfulness, 0.936 citation accuracy, 0.981 refusal correctness).
- Designed the agent control flow (entity detection pinning retrieval to the named company's filing, per-entity budgets, bounded agentic retry, prompt-injection and MNPI guardrails) and implemented it end to end.

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Architected and shipped a production multi-agent system (19 REST endpoints) orchestrated by a 3-layer LangGraph hierarchy (planning supervisor, parallel specialists, reflective reviewer), multi-model routing (GPT-4o, Claude Haiku) and two-tier Redis/pgvector memory, validated by 106/113 tests on a live deployment.
- Built a custom MCP server wiring internal and third-party tools into agents across 6 agent skills, with OpenTelemetry distributed tracing and a Human-in-the-Loop system spanning 4 escalation severity levels with audit-trail logging, built using spec-driven development with Claude Code.

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI & Machine Learning: Multi-Agent Orchestration, LangGraph, LangChain, Tool Orchestration (MCP), LLM Application Development, Retrieval-Augmented Generation, Prompt Engineering, Function Calling, Agent Evaluation (RAGAS), Guardrails, Multimodal AI, NLP, Embeddings, Vector Databases, PyTorch, TensorFlow

Infrastructure & DevOps: FastAPI, ChromaDB, Docker, OpenTelemetry (Observability), Google Cloud Platform, Azure, PostgreSQL, GitHub Actions, CI/CD, Git, REST APIs, ReactJS, OpenAI/Claude/Gemini APIs

EDUCATION

Stevens Institute of Technology

May 2026

Master of Science – Applied Artificial Intelligence

NJ, US

PES University

May 2024

Bachelor of Technology – Computer Science (Specialization in ML)

KA, IND

PUBLICATIONS AND CERTIFICATIONS

- Adaptive Ensemble Detection for Medical Imaging, M.S. Capstone, Stevens (Advised by Prof. Kevin Lu): a size-conditioned Weighted Box Fusion ensemble of Faster R-CNN, Mask R-CNN, and YOLO26 reaching 0.71 mAP@0.5 on the 2018 Data Science Bowl.
- Best Research Paper Award, ICICC 2024: WorkFit, a real-time multimodal posture monitoring system using CNN-based computer vision and flex sensor fusion.